

The circular chromatic number of the orthogonality graph of $\mathbb{R}^3$ is 4

A proof of Conjecture 4.41 of Ghebleh, answering a problem of DeVos, Ghebleh, Goddyn, Mohar and Naserasr

github.com/graph-theory-AI

Machine-generated note

17 September 2026

Abstract

Let $\mathcal{O}$ be the graph whose vertices are the lines through the origin of $\mathbb{R}^3$, two lines being adjacent when they are perpendicular. We prove that $\mathcal{O}$ admits no homomorphism to any *finite* graph in which every vertex neighbourhood induces a bipartite graph. Since the circular clique $K_{p/q}$ has this property exactly when $p < 4q$, it follows that the circular chromatic number of $\mathcal{O}$ is $\chi_c(\mathcal{O}) = 4$; this answers a problem of DeVos, Ghebleh, Goddyn, Mohar and Naserasr posed on the Open Problem Garden and proves Conjecture 4.41 of Ghebleh's 2007 thesis, where the best published lower bound, $\chi_c(\mathcal{O}) \geq 11/3$, was obtained by computer search. The proof rests on one elementary geometric lemma: for any three linearly independent vectors a, b, c , the lines contained in $a^\perp \cup b^\perp \cup c^\perp$ induce a non-bipartite subgraph of $\mathcal{O}$, an odd closed walk being produced by iterating cross products until a 2×2 matrix acquires a real eigenvector.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 6](#) records the full provenance.

1 Introduction

The graph. Throughout, $\mathcal{O}$ denotes the *orthogonality graph* of $\mathbb{R}^3$: its vertex set is the set of one-dimensional subspaces (lines through the origin) of $\mathbb{R}^3$, and two lines are adjacent when they are perpendicular. For a nonzero vector x we write $[x]$ for the line it spans. Adjacent lines are distinct, so $\mathcal{O}$ is a simple graph; it is infinite, indeed its vertex set is the real projective plane. Coloring properties of $\mathcal{O}$ are of interest in the foundations of quantum mechanics: Kochen and Specker [4] showed that $\mathcal{O}$, and even certain finite subgraphs of it, has no independent set meeting every triangle, which gives $\chi(\mathcal{O}) \geq 4$; a proper 4-colouring (credited in [2] to Godsil and Zaks [5]) shows $\chi(\mathcal{O}) = 4$.

Circular colourings. For a real $r \geq 1$, a *circular r -colouring* of a graph G assigns to each vertex a point of the circle $\mathbb{R}/r\mathbb{Z}$ of circumference r so that adjacent vertices are at circular distance at least 1. For integers $p \geq 2q > 0$ the *circular clique* $K_{p/q}$ has vertex set $\mathbb{Z}_p$ and

$$a \sim b \iff q \leq (b - a \bmod p) \leq p - q.$$

A homomorphism $G \rightarrow K_{p/q}$ is the same as a circular p/q -colouring using only the p colours $\frac{1}{q}\mathbb{Z}_p \subseteq \mathbb{R}/\frac{p}{q}\mathbb{Z}$. The circular chromatic number can be defined either as

$$\chi_c(G) = \inf\{r : G \text{ has a circular } r\text{-colouring}\} \quad \text{or as} \quad \chi_c(G) = \inf\{p/q : G \rightarrow K_{p/q}\},$$

and for finite graphs the two agree and the infimum is attained (see Zhu's survey [7]). For infinite graphs such as $\mathcal{O}$ the two agree as well, and the direction we need is proved in [Lemma 4.2](#): a circular r -colouring

with $r < 4$ yields a homomorphism to some $K_{p/q}$ with $p/q < 4$. In every case $\chi_c(G) \leq \chi(G)$, and $\chi_c(G) \geq 3$ when G contains a triangle. Hence $3 \leq \chi_c(\mathcal{O}) \leq 4$, and the question is which.

The problem. The following was posted on the Open Problem Garden on 23 September 2008.

Question 1.1 (DeVos, Ghebleh, Goddyn, Mohar, Naserasr [1]). Let $\mathcal{O}$ denote the graph with vertex set consisting of all lines through the origin in $\mathbb{R}^3$ and two vertices adjacent in $\mathcal{O}$ if they are perpendicular. Is $\chi_c(\mathcal{O}) = 4$?

The Open Problem Garden entry records the bound $\chi_c(\mathcal{O}) \geq 3.5$, obtained “by investigating certain finite subgraphs of $\mathcal{O}$ ”, and this is also the state of the art recorded in the catalog page from which the problem was taken.¹ Both are out of date, and the catalog’s guess at a reference is wrong; see [Section 1](#).

The origin of the conjecture: Ghebleh’s thesis. The LLM referee’s priority search ([Appendix B](#)) located the actual source. Section 4.4 of Ghebleh’s 2007 doctoral thesis [2], “The Projective Plane Orthogonality Graph”, is devoted to exactly the graph $\mathcal{O}$, with the Kochen–Specker motivation. Writing G_n for the subgraph of $\mathcal{O}$ induced by the primitive integer vectors of sup-norm at most n , the thesis proves $\chi(\mathcal{O}) = 4$ (Theorem 4.37), $\chi_c(\mathcal{O}) \geq 7/2$ via $\chi_c(G_1) = 7/2$ on the 13-vertex graph G_1 (Proposition 4.39) – this is the “3.5” of the Open Problem Garden entry – and, by computer, $\chi_c(\mathcal{O}) \geq 11/3$ via a 44-vertex subgraph $H_2 \subseteq G_2$ with $\chi_c(H_2) = 11/3$ (Theorem 4.40). It also exhibits a $(27, 7)$ -colouring of the 145-vertex graph G_3 , so that $11/3 \leq \chi_c(G_3) \leq 27/7$, and states:

Conjecture 1.2 (Ghebleh [2, Conjecture 4.41]). $\chi_c(\mathcal{O}) = 4$.

Thus the published record lower bound before this note was $11/3$, not $7/2$, and the statement proved below is a named published conjecture, posed a year before the Open Problem Garden entry. The catalog page attributes the bound 3.5 to the paper with DOI 10.1137/050639715, “Coloring an orthogonality graph”; that paper is Godsil and Newman [6] and concerns the *ordinary* chromatic number of the orthogonality graph on the vertices of the ± 1 hypercube, a different graph, and it is not the source of the bound. The related paper “Circular coloring the plane” of DeVos, Ebrahimi, Ghebleh, Goddyn, Mohar and Naserasr [3] (Section 4.3 of the thesis) concerns the unit-distance graph of the plane, not $\mathcal{O}$; its argument is a supremum over $\sqrt{3}$ -pairs with a $K_4 - e$ gadget and does not use the obstruction of this note, which could not work there since in the plane unit-distance graph every vertex neighbourhood induces a disjoint union of 6-cycles and is bipartite.

What is new. The odd-walk lemma ([Lemma 3.1](#)), the reduction of circular colourings below 4 to homomorphisms into finite locally bipartite graphs, and the resulting proof of [Conjecture 1.2](#) are, to the best of the referee’s search, not in the literature: Section 4.4 of [2] proceeds entirely by computation on the finite graphs G_n , and the referee found nothing else on $\chi_c(\mathcal{O})$ in zbMATH, OpenAlex, Crossref, Semantic Scholar or arXiv. Novelty is thus unrefuted rather than established; see [Remark 5.7](#).

Locally bipartite graphs. We call a graph H *locally bipartite* if for every vertex α the open neighbourhood $N_H(\alpha)$ induces a bipartite subgraph $H[N_H(\alpha)]$. (The same words are used by Illingworth [8] for an unrelated extremal question; the collision is terminological only.) For $S \subseteq V(\mathcal{O})$ we write $N_{\mathcal{O}}(S) = \bigcup_{L \in S} N_{\mathcal{O}}(L)$, and note that $N_{\mathcal{O}}([a])$ is precisely the set of lines contained in the plane $P_a := a^\perp$.

2 Main result

Theorem 2.1 (Main theorem). *There is no homomorphism from $\mathcal{O}$ to a finite locally bipartite graph.*

¹https://graph-theory-ai.github.io/graph-conjectures/op/circular_colouring_the_orthogonality_graph/

Equivalently, every finite homomorphic image of $\mathcal{O}$ has a vertex whose neighbourhood contains an odd cycle, i.e. contains an odd wheel. The word *finite* cannot be dropped: $\mathcal{O}$ is itself locally bipartite (Remark 5.1).

Corollary 2.2. $\chi_c(\mathcal{O}) = 4$, for either definition of the circular chromatic number. In particular the answer to Question 1.1 is yes, and Conjecture 1.2 holds.

Proof. By Lemma 4.1 the circular clique $K_{p/q}$ is locally bipartite whenever $p < 4q$, so by Theorem 2.1 there is no homomorphism $\mathcal{O} \rightarrow K_{p/q}$ with $p/q < 4$; this gives $\chi_c(\mathcal{O}) \geq 4$ for the homomorphism definition. If $\mathcal{O}$ had a circular r -colouring with $r < 4$, Lemma 4.2 would produce a homomorphism $\mathcal{O} \rightarrow K_{p/q}$ with $p/q < 4$, so $\chi_c(\mathcal{O}) \geq 4$ for the real-circle definition as well. The proper 4-colouring of Proposition 4.3 gives $\chi_c(\mathcal{O}) \leq \chi(\mathcal{O}) \leq 4$. $\square$

Idea of proof. Let $h: \mathcal{O} \rightarrow H$ be a homomorphism to a finite locally bipartite graph, and for $\alpha \in V(H)$ let $S_\alpha = h^{-1}(\alpha)$ be the fibre. Every neighbour of a line in S_α is mapped into $N_H(\alpha)$, so h restricts to a homomorphism $\mathcal{O}[N_\mathcal{O}(S_\alpha)] \rightarrow H[N_H(\alpha)]$; the target is bipartite, hence so is the source. The odd-walk lemma (Lemma 3.1) says that if S contains three linearly independent lines $[a], [b], [c]$ then $\mathcal{O}[N_\mathcal{O}(S)] \supseteq \mathcal{O}[\text{lines in } P_a \cup P_b \cup P_c]$ is *not* bipartite. So every fibre spans a subspace of dimension at most 2. But if $m = |V(H)|$, three of the $2m + 1$ moment-curve lines $[(1, j, j^2)]$, $1 \leq j \leq 2m + 1$, lie in a common fibre, and any three of them are linearly independent by the Vandermonde determinant. The only use of finiteness is this pigeonhole step. [\[full proof\]](#)

3 The odd-walk lemma

For a nonzero vector $a \in \mathbb{R}^3$ put $P_a = a^\perp$ and $J_a(x) = a \times x$. When $J_a(x) \neq 0$ the line $[J_a(x)]$ is perpendicular to $[x]$ and lies in P_a ; so each application of a cross product is one edge of $\mathcal{O}$, landing in a prescribed plane. The kernel of J_a is $\mathbb{R}a$, so the restriction $J_d: P_e \rightarrow P_d$ is injective whenever $d \cdot e \neq 0$.

Lemma 3.1 (Odd-walk lemma). *Let $a, b, c \in \mathbb{R}^3$ be linearly independent. The subgraph of $\mathcal{O}$ induced by the lines contained in $P_a \cup P_b \cup P_c$ is not bipartite.*

Idea of proof. If two of the normals are perpendicular, say $a \perp b$, then $[a], [b], [a \times b]$ is a triangle inside the union. Otherwise normalise so that $a = (0, 0, 1)$, $b = (s, 0, t)$ with $s > 0$, $0 < t < 1$, and $c = (u, v, w)$; linear independence and pairwise non-orthogonality mean $v \neq 0$, $w \neq 0$, $su + tw \neq 0$. Identify P_a with $\mathbb{R}^2$ and consider the two linear operators on P_a

$$E = J_a^2 J_b^2 = \begin{pmatrix} t^2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F = J_a J_c J_b = \begin{pmatrix} 0 & su + tw \\ -tw & sv \end{pmatrix}.$$

F is a 3-edge walk $P_a \rightarrow P_b \rightarrow P_c \rightarrow P_a$ and E is a 4-edge walk $P_a \rightarrow P_b \rightarrow P_b \rightarrow P_a \rightarrow P_a$, all of whose vertices lie in the union and none of whose vectors vanish, by the injectivity remark. The discriminant of the characteristic polynomial of $E^n F$ is $(sv)^2 - 4t^{2n} tw(su + tw) \rightarrow (sv)^2 > 0$, so for some n the invertible matrix $E^n F$ has a real eigenvector $x_0 \neq 0$. Starting at $[x_0]$ and performing the $3 + 4n$ cross-product steps returns to $[E^n F x_0] = [x_0]$: a closed walk of odd length.

In geometric language (the referee's): the lines of each plane form a projective line $\mathbb{RP}^1$, the cross products are projective-linear maps between these circles, and an odd cycle is a real fixed point of an odd-length composite. The 3-step composite F may be elliptic, with no real fixed point, so one pads it with powers of the hyperbolic 4-step composite E , which preserves parity and drives the discriminant positive. [\[full proof\]](#)

Corollary 3.2. *If $S \subseteq V(\mathcal{O})$ and $\mathcal{O}[N_\mathcal{O}(S)]$ is bipartite, then the lines in S span a subspace of $\mathbb{R}^3$ of dimension at most 2.*

Proof. Otherwise S contains lines $[a], [b], [c]$ with a, b, c linearly independent. The set of lines contained in $P_a \cup P_b \cup P_c$ is $N_\mathcal{O}([a]) \cup N_\mathcal{O}([b]) \cup N_\mathcal{O}([c]) \subseteq N_\mathcal{O}(S)$, so the non-bipartite graph of Lemma 3.1 is an induced subgraph of $\mathcal{O}[N_\mathcal{O}(S)]$. $\square$

4 Circular cliques and the four-colouring

Lemma 4.1. *Let $p \geq 2q > 0$. If $p < 4q$ then $K_{p/q}$ is locally bipartite.*

Idea of proof. By translation symmetry it suffices to treat $N(0) = \{q, \dots, p - q\}$. Since $p - q < 3q$, the two blocks $\{q, \dots, 2q - 1\}$ and $\{2q, \dots, 3q - 1\}$ cover $N(0)$, and any q consecutive residues are pairwise non-adjacent in $K_{p/q}$. [\[full proof\]](#)

The referee verified exhaustively for $q \leq 12$, $2q \leq p \leq 6q$ that $K_{p/q}$ is locally bipartite if and only if $p < 4q$, so the lemma is sharp; equivalently, χ_c of every odd wheel is 4 [\[7\]](#).

Lemma 4.2 (From real-circle to finite-palette colourings). *Let G be a graph, finite or infinite, with a circular r -colouring for some real r with $2 \leq r < 4$. Then there are integers $p \geq 2q > 0$ with $p/q < 4$ and a homomorphism $G \rightarrow K_{p/q}$.*

Idea of proof. Choose a rational R with $r < R < 4$ and rescale the colouring to circumference R ; every edge now has circular distance at least $R/r > 1$. Write $R = p/q$ with q so large that $2/q < R/r - 1$ (the fraction need not be in lowest terms) and round every colour to the grid $\frac{1}{q}\mathbb{Z}_p$. Each colour moves by less than $1/q$, so every edge keeps distance at least 1, which is exactly adjacency in $K_{p/q}$. The estimate is uniform in the vertex, so the number of vertices is irrelevant. [\[full proof\]](#)

Proposition 4.3 (Explicit proper 4-colouring). *Represent each line by the unit vector $u = (x, y, z)$ with $z > 0$, or $y > 0$ if $z = 0$, or $x > 0$ if $y = z = 0$. For $[u] \neq [e_3]$ the planar vector (x, y) is nonzero; colour $[u]$ by the half-open quadrant containing (x, y) , i.e. by which of the argument intervals $[0, \frac{\pi}{2})$, $[\frac{\pi}{2}, \pi)$, $[\pi, \frac{3\pi}{2})$, $[\frac{3\pi}{2}, 2\pi)$ contains $\arg(x, y)$; colour the pole $[e_3]$ with the third colour, that of $[\pi, \frac{3\pi}{2})$. This is a proper 4-colouring of $\mathcal{O}$; hence $\chi(\mathcal{O}) = 4$.*

Idea of proof. Two nonzero planar vectors in the same half-open quadrant have positive inner product, and all chosen representatives have $z \geq 0$, so two non-pole lines of the same colour have $u \cdot v = (x, y) \cdot (x', y') + zz' > 0$. Equatorial representatives have argument in $[0, \pi)$, so no equatorial line gets the third colour, and every other third-colour line has $z > 0$ and is not perpendicular to e_3 . [\[full proof\]](#)

The 4-colourability of $\mathcal{O}$ is of course classical [\[5, 2, 1\]](#); the octahedron colouring of [\[1\]](#) leaves a choice on the boundaries, and the point of [Proposition 4.3](#) is only to fix one explicitly.

Proposition 4.4 (Finite witnesses). *For every integer $m \geq 1$ there is a finite induced subgraph $F_m \subseteq \mathcal{O}$ with no homomorphism to any locally bipartite graph on at most m vertices; in particular $F_m \not\rightarrow K_{p/q}$ for every $p \leq m$ with $p/q < 4$. Explicitly, F_m is induced by the $2m + 1$ lines $[(1, j, j^2)]$, $1 \leq j \leq 2m + 1$, together with, for each triple of them, the lines of the odd closed walk supplied by [Lemma 3.1](#) (the triangle $[a], [b], [a \times b]$ when two of the three normals are perpendicular, and otherwise the walk of length $3 + 4n$ for the least $n \geq 0$ with positive discriminant).*

Idea of proof. This is the finite shadow of the proof of [Theorem 2.1](#): under a homomorphism to a locally bipartite graph on at most m vertices, three of the $2m + 1$ initial lines share a fibre, and their odd walk lies in the neighbourhood of that fibre, which must map into a bipartite neighbourhood. [\[full proof\]](#)

5 Remarks

Remark 5.1 (The finiteness hypothesis is essential). The graph $\mathcal{O}$ is itself locally bipartite: $N_{\mathcal{O}}(L)$ is the set of lines of the plane $L^{\perp}$, and within that plane every line has exactly one perpendicular line, so $\mathcal{O}[N_{\mathcal{O}}(L)]$ is a perfect matching. Since the identity $\mathcal{O} \rightarrow \mathcal{O}$ is a homomorphism, [Theorem 2.1](#) is false without the word *finite*, and the pigeonhole step of its proof is genuinely load-bearing. The original writeup states the hypothesis correctly but does not comment on it; the referee checked the claim on the 577 primitive integer directions of sup-norm at most 5 (every neighbourhood bipartite, maximum degree 1 inside each neighbourhood). Relatedly, the theorem does *not* say that some finite subgraph of $\mathcal{O}$ has circular chromatic number 4; the opposite is true, see [Remark 5.3](#).

Remark 5.2 (Size of the witnesses). In [Lemma 3.1](#) the least admissible n is unbounded. With the notation of the proof, $\Delta_n = (sv)^2 - 4t^{2n+1}w(su + tw)$, so $n = 0$ works when $w(su + tw) \leq 0$, and otherwise the least n is the least integer with $2n + 1 > \log((sv)^2/(4w(su + tw)))/\log t$, which blows up as $a \cdot b \rightarrow \pm 1$ (nearly parallel normals) or as $sv \rightarrow 0$. The referee’s construction of F_7 (the case of the smallest palette $K_{7/2}$) from the 15 moment-curve lines processed all $\binom{15}{3} = 455$ triples: 311 walks were fully built with zero violations, of lengths 39 to 11 927 (median 639), and 144 triples needed $n > 3000$, the largest being $n = 1\,037\,597$, a walk of about 4.15 million edges; so $|V(F_7)|$ is of order 10^7 . By contrast the first attempt’s [Proposition 2](#) ([Remark 5.3](#)) shows that at least 11 vertices are needed to block $K_{7/2}$, and Ghebleh’s 49-vertex G_2 already does so. The moment curve is a poor choice of starting lines for a small witness; any $2m + 1$ lines in general position would do, and the writeup’s phrase “a terminating search for n ” undersells how large F_m becomes. This is a size issue, not a soundness issue.

Remark 5.3 (Relation to the first attempt and to the earlier finite computations). This result is the second attack on the problem within the campaign ([Section 6](#)). The first, by the same model, reported only partial results, which the referee of that writeup checked and confirmed: *Proposition 1*, if $\chi_c(\mathcal{O}) < 4$ then $\chi_c(\mathcal{O}) = 4 - 1/k$ for an integer $k \geq 2$ and $\mathcal{O} \rightarrow K_{(4k-1)/k}$ (from the facts that any two lines have a common perpendicular and that an optimal colouring of a finite graph contains a regular p -gon of colours); and *Proposition 2*, every finite subgraph $F \subseteq \mathcal{O}$ satisfies $\chi_c(F) < 4$, indeed $\chi_c(F) \leq 4 - 1/\lfloor (|V(F)| + 1)/4 \rfloor$, by projecting the lines to a generic plane. Neither is used in the present proof. [Proposition 2](#) is consistent with [Proposition 4.4](#): F_m does map to some $K_{p/q}$ with $p/q < 4$, but only with $p > m$.

That earlier referee report (verification_astra/circular_colouring_the_orthogonality_graph.OPG-PARTIAL.md) also ran exact SAT computations on the subgraphs $\mathcal{O}_n$ of $\mathcal{O}$ induced by the primitive integer directions of sup-norm at most n (these are Ghebleh’s G_n): $\mathcal{O}_2$ (49 lines, 138 edges) has $\chi_c(\mathcal{O}_2) = 11/3$; $\mathcal{O}_3$ (145 lines, 546 edges) admits no homomorphism to $K_{11/3}$ or $K_{15/4}$; and $\mathcal{O}_4$, the 289 lines of sup-norm at most 4 (1326 edges), admits no homomorphism to $K_{19/5}$, while $\mathcal{O}_4 \rightarrow K_4$. This gave $\chi_c(\mathcal{O}) > 19/5$ unconditionally and, combined with [Proposition 1](#), $\chi_c(\mathcal{O}) \geq 23/6$, already above Ghebleh’s published $11/3$. That referee observed the monotone pattern ($\mathcal{O}_2$ excludes $7/2$, $\mathcal{O}_3$ excludes $15/4$, $\mathcal{O}_4$ excludes $19/5$) and named a uniform construction realising it for every k as the missing ingredient; [Proposition 4.4](#) is that construction, and [Corollary 2.2](#) supersedes all of these bounds.

Remark 5.4 (The referee’s computational checks). The referee of the present writeup ([Appendix B](#)) verified every finite object in code, the core lemma in exact arithmetic.

1. *Odd-walk lemma, exactly.* For seven explicit integer triples (a, b, c) the matrices E, F were computed as exact integer matrices in the basis $(a \times b, a \times (a \times b))$ of P_a , and the eigenvector and the whole walk were carried out in the quadratic field $\mathbb{Q}(\sqrt{\Delta_n})$ with no floating point; every orthogonality, non-vanishing, plane membership and the projective closure held as exact identities. Walk lengths: 3 (the perpendicular case), 63, 135, 59, 3, 19, 3; the two length-3 walks for pairwise non-orthogonal normals show that $n = 0$ can occur even then.
2. *Odd-walk lemma, randomly.* 3988 random triples were fully built (three exceeded a cap of $n \leq 4000$) with zero failures; every walk had length exactly $3 + 4n$, 2097 of them triangles, the longest 9607 edges. A stress test with $a \cdot b \rightarrow 1$ gave 67 further successes, longest walk 15 183.
3. *Local bipartiteness of $K_{p/q}$.* Exhaustive for $q \leq 12$, $2q \leq p \leq 6q$: locally bipartite iff $p < 4q$, with the writeup’s blocks A, B checked to cover $N(0)$ and to be independent. Cross-checked from the other side with an independent exact χ_c solver: $\chi_c(W_3) = \chi_c(W_5) = \chi_c(W_7) = \chi_c(W_9) = 4$.
4. *The 4-colouring.* On the 865 primitive integer directions of sup-norm at most 6 (5502 exactly orthogonal pairs, integer arithmetic): 0 monochromatic edges, class sizes 228/228/205/204, the 48 equatorial lines using only the first two colours and the pole the third; also 0 monochromatic among 200 000 random exactly perpendicular real pairs.
5. *Ghebleh’s data.* His G_1, G_2, G_3 were rebuilt from scratch (13/24, 49/138, 145/546 vertices/edges, matching the thesis) and the homomorphism questions re-run with z3, calibrated on eight clique-

to-clique instances: $G_2 \nrightarrow K_{7/2}$, $G_3 \nrightarrow K_{7/2}$, $K_{11/3}$, $G_3 \rightarrow K_{27/7}$ (every satisfying colouring re-verified edge by edge). This reproduces Theorem 4.40 of [2] and his $(27, 7)$ -colouring, and shows $11/3 < \chi_c(G_3) \leq 27/7$. The existence of $G_3 \rightarrow K_{27/7}$ is a genuine test the theorem passes: G_3 is finite and contains none of the (generally irrational) lines of the odd walks.

No computational check failed.

Remark 5.5 (A natural objection that does not apply). $K_{7/2}$ is locally bipartite, yet for its triangle $\{0, 2, 4\}$ the union $N(0) \cup N(2) \cup N(4)$ is all of $\mathbb{Z}_7$ and is not bipartite; so one might think an odd walk inside a union of three neighbourhoods cannot obstruct a homomorphism to $K_{7/2}$. The argument is not of that form: the three lines $[a]$, $[b]$, $[c]$ are taken in a single fibre $h^{-1}(\alpha)$, so the union of their neighbourhoods maps into the single neighbourhood $N_H(\alpha)$, which is bipartite by hypothesis. The pigeonhole is applied to fibres, not to triangles.

Remark 5.6 (Minor points in the writeup, left as they are). (i) In [Lemma 4.2](#) the writeup’s constant $2/q$ and the phrase “changes each colour by less than $1/q$ ” are safe over-estimates: nearest-grid rounding moves a colour by at most $1/(2q)$ and a distance by at most $1/q$. (ii) Writing $R = p/q$ “choosing a sufficiently large common multiple” means p/q is not in lowest terms, which is harmless since $K_{p/q}$ is defined for all $p \geq 2q$. (iii) For $p/q < 2$ the circular clique is undefined, but $\mathcal{O}$ contains triangles so $\chi_c(\mathcal{O}) \geq 3$ anyway. (iv) A loop in the target H is excluded automatically, since a loop at α makes $H[N_H(\alpha)]$ non-bipartite; and the case $V(H) = \emptyset$ is trivial.

Remark 5.7 (Novelty). Correctness and priority are separate. The writeup’s own caveat field says “literature priority has not been checked”. The referee searched zbMATH Open, OpenAlex (including full text), Crossref, Semantic Scholar and arXiv, the citers of [3], Ghebleh’s complete publication list (Section 4.4 of the thesis was never turned into a paper), and the Open Problem Garden entry itself (still listed open, bibliography empty, last comment 2009), and found no proof of $\chi_c(\mathcal{O}) = 4$ and no improvement on $11/3$; but no general web search engine was reachable from its session, so an unindexed preprint on a personal page would have been missed. The result should be regarded as correct and apparently new, with novelty unconfirmed.

6 Provenance

The mathematics in this note was produced without human intervention, in three stages.

(1) The argument was found by the language model gpt-6-astra (reasoning effort *max*) in a single autonomous pass started on 2026-09-16, as part of the campaign’s `attacks_retry` leg (catalog id `circular_colouring_the_orthogonality_graph`, an OpenProblemGarden problem; original writeup in `attacks_retry/circular_colouring_the_orthogonality_graph/output.md`). This was a *second* attempt: the prompt contained, as unverified context, the first attempt by the same model in the `attacks_opg` leg, which had reported only the partial results summarised in [Remark 5.3](#) (no referee report on that attempt was supplied). The new writeup discards the first attempt’s discrete-spectrum route and introduces the odd-walk lemma; the material in [Remark 5.3](#) attributed to the first attempt is inherited and is not claimed as new here.

(2) An independent adversarial referee agent (claude-opus-5, 2026-09-17) re-derived every step, found none to be a gap or an error, ran the computational checks listed in [Remark 5.4](#) (the odd-walk lemma in exact arithmetic over a real quadratic field; exhaustive verification of [Lemma 4.1](#) and its sharpness; the 4-colouring on 865 integer directions; reconstruction of Ghebleh’s G_1, G_2, G_3 and re-derivation of his Theorem 4.40 by SAT), carried out the priority search that located [2], and returned the verdict `CONFIRMED` with high confidence, the sole reservation being that novelty is unrefuted rather than established. Its report is reproduced verbatim in [Appendix B](#). A separate, earlier referee review of the weaker first attempt (`verification_astra/circular_colouring_the_orthogonality_graph.OPG-PARTIAL.md`) is summarised in [Remark 5.3](#).

(3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The text was reorganised, with proof ideas in the main text and full proofs in [Appendix A](#). Every deviation from

the writeup is listed here.

- *No repairs.* No mathematical step was changed; the referee found none needing repair. The statements of [Theorem 2.1](#), [Lemma 3.1](#), [Lemma 4.1](#), [Lemma 4.2](#) and [Proposition 4.3](#) are those of the writeup (the writeup states the last three inline rather than as numbered results).
- *Clarification in the proof of [Lemma 3.1](#).* The writeup’s normalisation “choose their signs and an orthonormal coordinate system” is spelled out (flip the sign of b so that $t = a \cdot b > 0$, rotate about a so that $s \geq 0$), following the referee’s step L2.
- *Clarification in [Proposition 4.4](#).* The writeup’s finite-witness construction applies the E, F construction to every triple; triples with two perpendicular normals must instead use the triangle $[a], [b], [a \times b]$, as in the first case of the lemma. Added following the referee’s caveat 4; the proposition is otherwise the writeup’s Section 5.
- *Added remarks, all taken from the referee report:* [Remark 5.1](#) (finiteness essential, $\mathcal{O}$ locally bipartite), [Remark 5.2](#) (the explicit formula for the least n and the measured sizes), [Remark 5.5](#), [Remark 5.6](#), [Remark 5.4](#), [Remark 5.7](#), and the odd-wheel reformulation after [Theorem 2.1](#). The “projective holonomy” description in the idea of proof of [Lemma 3.1](#) is the referee’s paraphrase, so labelled.
- *Added citations and corrections of the record.* The writeup cites nothing. Added: Ghebleh’s thesis [2] as the origin of [Conjecture 1.2](#) and of the true prior lower bound $11/3$; [3]; [4], [5] for the classical facts $\chi(\mathcal{O}) = 4$; [6] to identify the catalog’s mis-citation; [7] for standard facts on χ_c ; [8] for the terminology collision. The statement that the catalog’s recorded state of the art ($7/2$) was wrong is the referee’s finding.
- *Context from the first attempt ([Remark 5.3](#))* is stated with attribution and without proof; it plays no role in the proof of [Corollary 2.2](#).

No human mathematician has checked this argument. We circulate it for human review.

A Full proofs

Lemma 3.1 (Odd-walk lemma). *Let $a, b, c \in \mathbb{R}^3$ be linearly independent. The subgraph of $\mathcal{O}$ induced by the lines contained in $P_a \cup P_b \cup P_c$ is not bipartite.*

Proof. Case 1: two of the normals are perpendicular, say $a \perp b$. Then $[a] \subseteq P_b$, $[b] \subseteq P_a$ and $[a \times b] \subseteq P_a \cap P_b$, so all three lines lie in $P_a \cup P_b \cup P_c$, and they are pairwise perpendicular: a triangle. (The plane P_c is not needed in this case.)

Case 2: a, b, c are pairwise non-orthogonal. Replace a, b, c by unit vectors; this changes none of the planes. Since $a \cdot b \neq 0$ we may replace b by $-b$ if necessary so that $t := a \cdot b > 0$, and since b is not parallel to a (linear independence) we have $0 < t < 1$. Choose an orthonormal coordinate system with third axis a and first axis along the component of b orthogonal to a (a rotation about a). Then

$$a = (0, 0, 1), \quad b = (s, 0, t), \quad c = (u, v, w), \quad s > 0, \quad 0 < t < 1, \quad s^2 + t^2 = 1.$$

Since a, b span the xz -plane, linear independence of a, b, c is equivalent to $v \neq 0$; and $a \cdot c = w \neq 0$, $b \cdot c = su + tw \neq 0$ by pairwise non-orthogonality. Thus

$$v \neq 0, \quad w \neq 0, \quad su + tw \neq 0. \tag{1}$$

Non-vanishing. If $d \cdot e \neq 0$ then $J_d: P_e \rightarrow P_d$ is injective, because $\ker J_d = \mathbb{R}d$ and $\mathbb{R}d \cap P_e = \{0\}$ exactly when $d \notin P_e$. The transitions used below are $J_b: P_a \rightarrow P_b$ and $J_a: P_b \rightarrow P_a$ (need $a \cdot b \neq 0$), $J_c: P_b \rightarrow P_c$ (need $b \cdot c \neq 0$), $J_a: P_c \rightarrow P_a$ (need $a \cdot c \neq 0$), and $J_b: P_b \rightarrow P_b$, $J_a: P_a \rightarrow P_a$ (need $b \cdot b, a \cdot a \neq 0$). All hold, so no vector in any walk below is zero.

The two operators. Identify P_a with $\mathbb{R}^2$ via $(x_1, x_2, 0) \leftrightarrow (x_1, x_2)$. For $x = (x_1, x_2, 0)$ one computes

$$\begin{aligned} J_b(x) &= b \times x = (-tx_2, tx_1, sx_2), \\ J_b^2(x) &= b \times (-tx_2, tx_1, sx_2) = (-t^2x_1, -(s^2 + t^2)x_2, stx_1) = (-t^2x_1, -x_2, stx_1), \\ J_a(y) &= a \times y = (-y_2, y_1, 0) \quad \text{for any } y = (y_1, y_2, y_3), \end{aligned}$$

so $J_a^2 J_b^2(x) = (t^2 x_1, x_2, 0)$, and

$$\begin{aligned} J_c J_b(x) &= c \times (-tx_2, tx_1, sx_2) = (svx_2 - twx_1, -(su + tw)x_2, t(ux_1 + vx_2)), \\ J_a J_c J_b(x) &= ((su + tw)x_2, svx_2 - twx_1, 0). \end{aligned}$$

Hence, as operators on $P_a \cong \mathbb{R}^2$,

$$E := J_a^2 J_b^2 = \begin{pmatrix} t^2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F := J_a J_c J_b = \begin{pmatrix} 0 & su + tw \\ -tw & sv \end{pmatrix}. \quad (2)$$

(Operators compose right to left: F applies J_b first.) These have a walk interpretation. Applying F to a nonzero $x \in P_a$ visits the lines

$$[x] \subseteq P_a, \quad [J_b x] \subseteq P_b, \quad [J_c J_b x] \subseteq P_c, \quad [Fx] \subseteq P_a,$$

consecutive lines being perpendicular: a walk of 3 edges. Applying E visits

$$[x] \subseteq P_a, \quad [J_b x] \subseteq P_b, \quad [J_b^2 x] \subseteq P_b, \quad [J_a J_b^2 x] \subseteq P_a, \quad [Ex] \subseteq P_a,$$

a walk of 4 edges. (The step J_b applied to a vector already in P_b is the rotation of P_b by a right angle; it is legitimate because $b \notin P_b$.) Every line visited lies in $P_a \cup P_b \cup P_c$ and every vector is nonzero by the non-vanishing paragraph.

A real eigenvector. For $n \geq 0$, $E^n = \text{diag}(t^{2n}, 1)$ and

$$E^n F = \begin{pmatrix} 0 & t^{2n}(su + tw) \\ -tw & sv \end{pmatrix}, \quad \text{tr}(E^n F) = sv, \quad \det(E^n F) = t^{2n} tw(su + tw).$$

The discriminant of its characteristic polynomial is

$$\Delta_n = (sv)^2 - 4t^{2n} tw(su + tw). \quad (3)$$

Since $0 < t < 1$, $t^{2n} \rightarrow 0$, and since $s > 0$, $v \neq 0$ we have $(sv)^2 > 0$; hence $\Delta_n \rightarrow (sv)^2 > 0$ and we may fix $n \geq 0$ with $\Delta_n > 0$. Then $E^n F$ has two distinct real eigenvalues and in particular a real eigenvector $x_0 \neq 0$ in P_a , $E^n F x_0 = \lambda x_0$. By (1) and $t > 0$, $\det(E^n F) \neq 0$, so $\lambda \neq 0$.

The odd closed walk. Start at $[x_0]$, perform the three steps of F , then n times the four steps of E . This is a walk in $\mathcal{O}$ of length $3 + 4n$ all of whose vertices are lines in $P_a \cup P_b \cup P_c$, and it ends at $[E^n F x_0] = [\lambda x_0] = [x_0]$ since $\lambda \neq 0$ (working with lines rather than vectors is what makes a negative λ harmless). A closed walk of odd length in a simple graph contains an odd cycle, so the induced subgraph is not bipartite. $\square$

Theorem 2.1 (Main theorem). *There is no homomorphism from $\mathcal{O}$ to a finite locally bipartite graph.*

Proof. Suppose $h: \mathcal{O} \rightarrow H$ is a homomorphism to a finite locally bipartite graph H , and put $m = |V(H)|$. (If $m = 0$ there is no map from the non-empty $V(\mathcal{O})$; if H had a loop at α then $\alpha \in N_H(\alpha)$ and $H[N_H(\alpha)]$ would not be bipartite, so H is loopless.)

For $\alpha \in V(H)$ let $S_\alpha = h^{-1}(\alpha)$. If $L \in N_{\mathcal{O}}(S_\alpha)$ then L is adjacent to some $M \in S_\alpha$, so $h(L)$ is adjacent to $h(M) = \alpha$, i.e. $h(L) \in N_H(\alpha)$; and h maps edges to edges. Hence h restricts to a homomorphism

$$\mathcal{O}[N_{\mathcal{O}}(S_\alpha)] \rightarrow H[N_H(\alpha)].$$

The target is bipartite by hypothesis, and a graph admitting a homomorphism to a bipartite graph is bipartite (compose with the 2-colouring). So $\mathcal{O}[N_{\mathcal{O}}(S_\alpha)]$ is bipartite, and by Corollary 3.2 the lines in S_α span a subspace of dimension at most 2.

Now consider the $2m + 1$ distinct lines

$$L_j = [(1, j, j^2)], \quad 1 \leq j \leq 2m + 1.$$

Since $2m + 1 > 2m$, some fibre S_α contains three of them, say L_i, L_j, L_k with $i < j < k$. But

$$\det \begin{pmatrix} 1 & i & i^2 \\ 1 & j & j^2 \\ 1 & k & k^2 \end{pmatrix} = (j-i)(k-i)(k-j) \neq 0,$$

so $(1, i, i^2), (1, j, j^2), (1, k, k^2)$ are linearly independent and S_α spans $\mathbb{R}^3$, a contradiction. Hence no such h exists. $\square$

Lemma 4.1. *Let $p \geq 2q > 0$. If $p < 4q$ then $K_{p/q}$ is locally bipartite.*

Proof. The map $x \mapsto x + 1$ is an automorphism of $K_{p/q}$, so it suffices to show that $N(0)$ induces a bipartite graph. By definition $N(0) = \{q, q+1, \dots, p-q\}$. Put

$$A = N(0) \cap \{q, \dots, 2q-1\}, \quad B = N(0) \cap \{2q, \dots, 3q-1\}.$$

Since $p < 4q$ we have $p-q < 3q$, so $A \cup B = N(0)$. Each of A, B lies in a block of q consecutive residues; two distinct elements $x < y$ of such a block have $y-x \in \{1, \dots, q-1\}$ and $x-y \bmod p \in \{p-q+1, \dots, p-1\}$, neither of which lies in $[q, p-q]$ (using $p \geq 2q$), so they are non-adjacent. Thus A and B are independent sets covering $N(0)$, which is therefore bipartite. $\square$

Lemma 4.2 (From real-circle to finite-palette colourings). *Let G be a graph, finite or infinite, with a circular r -colouring for some real r with $2 \leq r < 4$. Then there are integers $p \geq 2q > 0$ with $p/q < 4$ and a homomorphism $G \rightarrow K_{p/q}$.*

Proof. Let $f: V(G) \rightarrow \mathbb{R}/r\mathbb{Z}$ be a circular r -colouring, $2 \leq r < 4$. Choose a rational number R with $r < R < 4$ and let $f_R = (R/r)f$, a map to $\mathbb{R}/R\mathbb{Z}$ under which every edge has circular distance at least $R/r > 1$. Write $R = p/q$ with integers $p, q > 0$, taking q (and hence $p = Rq$) large enough that

$$\frac{2}{q} < \frac{R}{r} - 1;$$

$p \geq 2q$ since $R \geq r \geq 2$, and $p/q = R < 4$. Let $g(v)$ be the point of the grid $\frac{1}{q}\mathbb{Z}/R\mathbb{Z} \cong \frac{1}{q}\mathbb{Z}_p$ nearest to $f_R(v)$ (any fixed tie-breaking rule); then $g(v)$ differs from $f_R(v)$ by less than $1/q$. For an edge uv the circular distance between $g(u)$ and $g(v)$ is therefore at least $R/r - 2/q > 1$. Multiplying by q identifies the grid with $\mathbb{Z}_p$, and a circular distance of at least 1 on the circle of circumference p/q between grid points means that their difference in $\mathbb{Z}_p$ lies in $[q, p-q]$, which is adjacency in $K_{p/q}$. So $gg: V(G) \rightarrow \mathbb{Z}_p$ is a homomorphism $G \rightarrow K_{p/q}$. The bound on the rounding error does not depend on the vertex, so no finiteness of $V(G)$ is needed. $\square$

Proposition 4.3 (Explicit proper 4-colouring). *Represent each line by the unit vector $u = (x, y, z)$ with $z > 0$, or $y > 0$ if $z = 0$, or $x > 0$ if $y = z = 0$. For $[u] \neq [e_3]$ the planar vector (x, y) is nonzero; colour $[u]$ by the half-open quadrant containing (x, y) , i.e. by which of the argument intervals $[0, \frac{\pi}{2})$, $[\frac{\pi}{2}, \pi)$, $[\pi, \frac{3\pi}{2})$, $[\frac{3\pi}{2}, 2\pi)$ contains $\arg(x, y)$; colour the pole $[e_3]$ with the third colour, that of $[\pi, \frac{3\pi}{2})$. This is a proper 4-colouring of $\mathcal{O}$; hence $\chi(\mathcal{O}) = 4$.*

Proof. Each line has exactly one unit representative $u = (x, y, z)$ obeying the stated rule, and $z \geq 0$ for all of them. For $[u] \neq [e_3]$ the vector (x, y) is nonzero, so its argument lies in exactly one of the four half-open intervals and the colouring is well defined. Number the colours 1, 2, 3, 4 in the order of the intervals.

Let $[u] \neq [v]$ be two non-pole lines of the same colour, with representatives $u = (x, y, z)$, $v = (x', y', z')$. The planar vectors (x, y) and (x', y') are nonzero and lie in the same half-open quadrant, so the angle between them is strictly less than $\pi/2$ and $(x, y) \cdot (x', y') > 0$. Since $z, z' \geq 0$,

$$u \cdot v = (x, y) \cdot (x', y') + zz' > 0,$$

so $[u]$ and $[v]$ are not perpendicular.

It remains to consider the pole $[e_3]$, of colour 3. Its neighbours are the equatorial lines ($z = 0$), whose representatives have $y > 0$, or $y = 0$ and $x > 0$; their arguments lie in $[0, \pi)$, so equatorial lines have colour 1 or 2, never 3. Every non-pole line of colour 3 has $z > 0$ and so is not perpendicular to e_3 . Hence no edge is monochromatic, $\chi(\mathcal{O}) \leq 4$, and $\chi(\mathcal{O}) \geq 4$ because $\chi_c(\mathcal{O}) \geq 4$ by [Corollary 2.2](#) (or by Kochen–Specker [\[4\]](#)). $\square$

Proposition 4.4 (Finite witnesses). *For every integer $m \geq 1$ there is a finite induced subgraph $F_m \subseteq \mathcal{O}$ with no homomorphism to any locally bipartite graph on at most m vertices; in particular $F_m \not\rightarrow K_{p/q}$ for every $p \leq m$ with $p/q < 4$. Explicitly, F_m is induced by the $2m + 1$ lines $[(1, j, j^2)]$, $1 \leq j \leq 2m + 1$, together with, for each triple of them, the lines of the odd closed walk supplied by [Lemma 3.1](#) (the triangle $[a], [b], [a \times b]$ when two of the three normals are perpendicular, and otherwise the walk of length $3 + 4n$ for the least $n \geq 0$ with positive discriminant).*

Proof. Fix $m \geq 1$ and let $T = \{[(1, j, j^2)] : 1 \leq j \leq 2m + 1\}$. For each 3-subset $\{[a], [b], [c]\} \subseteq T$ the vectors a, b, c are linearly independent (Vandermonde, as in the proof of [Theorem 2.1](#)), so [Lemma 3.1](#) applies: if two of a, b, c are perpendicular take the triangle $[a], [b], [a \times b]$; otherwise take the least $n \geq 0$ with $\Delta_n > 0$ in [\(3\)](#) — it exists since $\Delta_n \rightarrow (sv)^2 > 0$ — an eigenline $[x_0]$ of $E^n F$, and the closed walk of length $3 + 4n$ through $[x_0]$. Let W be the set of all lines on all these walks and triangles, and let $F_m = \mathcal{O}[T \cup W]$, a finite induced subgraph of $\mathcal{O}$. Its construction uses only field operations, one square root per triple (for x_0) and a terminating search for n .

Suppose $h: F_m \rightarrow H$ is a homomorphism to a locally bipartite graph with $|V(H)| \leq m$. Three lines $[a], [b], [c] \in T$ have the same image α . Every line of their odd closed walk lies in $P_a \cup P_b \cup P_c$, i.e. is adjacent in F_m to one of $[a], [b], [c]$, hence is mapped by h into $N_H(\alpha)$; and the walk’s edges are edges of F_m . So the odd closed walk maps to a closed walk of odd length in $H[N_H(\alpha)]$, which is bipartite: a contradiction. In particular $F_m \not\rightarrow K_{p/q}$ whenever $p \leq m$ and $p/q < 4$, by [Lemma 4.1](#). $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup (attacks_retry/circular_colouring_the_orthogonality_graph/output.md). Its step labels (L*, C, H*, K*, U*, W*) refer to that writeup’s sections, not to this note. The scripts it mentions are in verification_astra/scripts/circular_colouring_the_orthogonality_graph/. Treat it as machine output, not as an authority.

B.1 Interpretation

The problem (OpenProblemGarden, DeVos–Ghebleh–Goddyn–Mohar–Naserasr, 2008) asks whether $\chi_c(\mathcal{O}) = 4$ for the **continuous** orthogonality graph $\mathcal{O}$: vertices are all lines through the origin in $\mathbb{R}^3$, adjacent when perpendicular. The writeup opens with exactly this object (“the one-dimensional subspaces of $\mathbb{R}^3$, with adjacency given by orthogonality”) and never leaves it. There is no drift into the ± 1 -hypercube orthogonality graph of Frankl–Rödl / Godsil–Newman, no substitution of a finite subgraph for $\mathcal{O}$, and no weakening of the question.

Both standard definitions of χ_c for an infinite graph are handled: the homomorphism (finite-palette) definition $\chi_c(G) = \inf\{p/q : G \rightarrow K_{p/q}\}$ in §3, and the real-circle definition (colours in $\mathbb{R}/r\mathbb{Z}$, adjacent vertices at circular distance ≥ 1) in the second half of §3. The answer claimed — $\chi_c(\mathcal{O}) = 4$, i.e. **yes** to the problem — is the intended reading, and the author of the original problem would consider it resolved. There is no “as literally stated” gaming and no strawman.

The writeup in fact proves something strictly stronger than asked:

Theorem. $\mathcal{O}$ has no homomorphism to any *finite* graph in which every open neighbourhood induces a bipartite graph (“locally bipartite”).

$\chi_c(\mathcal{O}) \geq 4$ is the corollary, because $K_{p/q}$ is locally bipartite exactly when $p < 4q$. Equivalently: **every finite homomorphic image of $\mathcal{O}$ contains an odd wheel**. The one-line verdict is an accurate, if slightly understated, description of what is proved.

Interpretation: fair.

B.2 Step-by-step findings

I number the steps L* (§1 lemma), C (corollary), H* (§2), K* (§3), U* (§4), W* (§5).

step	label	note
L0. Setup: $P_a = a^\perp$, $J_a(x) = a \times x$; $J_a(x) \neq 0$ represents a line $\perp [x]$ lying in P_a	VALID	Immediate: $(a \times x) \cdot x = 0$ and $(a \times x) \cdot a = 0$. Adjacency in $\mathcal{O}$ is exactly perpendicularity of the two lines, and perpendicular lines are automatically distinct, so no loop can be created.
L1. Case “two normals perpendicular”, say $a \perp b$: $[a], [b], [a \times b]$ is a triangle inside $P_a \cup P_b \cup P_c$	VALID	$[a] \subseteq P_b$ and $[b] \subseteq P_a$ because $a \perp b$; $[a \times b] \subseteq P_a \cap P_b$. All three pairwise perpendicular. Verified exactly in code for $([1,0,0], [0,1,0], [1,2,3])$ — the triangle is the standard basis. Note this case needs only two of the three planes; c is not used, which is fine.
L2. Normalisation $a = (0,0,1)$, $b = (s,0,t)$ with $s > 0$, $0 < t < 1$, and $c = (u,v,w)$	VALID	Unit-normalise; $t = a \cdot b \neq 0$ by non-orthogonality, flip the sign of b to get $t > 0$; rotate about a to get $s \geq 0$, and $s > 0$ because $b \not\parallel a$; then $t < 1$. The sign/rotation choices are stated only as “choose their signs and an orthonormal coordinate system”, but they are exactly the two just described and change nothing projectively.
L3. Condition (1): $v \neq 0$, $w \neq 0$, $su + tw \neq 0$	VALID	a, b span the xz -plane, so linear independence of $a, b, c \iff v \neq 0$. $w = a \cdot c \neq 0$ and $su + tw = b \cdot c \neq 0$ are the two remaining non-orthogonality hypotheses.
L4. Injectivity: $J_d : P_e \rightarrow P_d$ is injective whenever $d \cdot e \neq 0$	VALID	$\ker J_d = \text{Rd}$, and $\text{Rd} \cap P_e = 0 \iff d \notin P_e \iff d \cdot e \neq 0$. The blanket claim “this applies to every transition below” is true: the transitions actually used are $P_a \rightarrow P_b$ (needs $a \cdot b \neq 0$), $P_b \rightarrow P_a$ (need $b \cdot a \neq 0$), $P_b \rightarrow P_c$ ($b \cdot c \neq 0$), $P_c \rightarrow P_b$ ($c \cdot b \neq 0$), $P_b \rightarrow P_a$ ($a \cdot b \neq 0$). All are covered by pairwise non-orthogonality plus
L5. Matrix formulas (2): $E = J_a^2 J_b^2 = \text{diag}(t^2, 1)$ and $F = J_a J_c J_b = [[0, su+tw], [-tw, sv]]$ on P_a	VALID	Re-derived by hand and confirmed symbolically. $J_b(x_1, x_2, 0) = (-tx_2, tx_1, sx_2)$; $J_b^2 x = (-t^2 x_1, -x_2, stx_1)$; $J_a(y) = (-y_2, y_1, 0)$, so $J_a^2 J_b^2 x = (t^2 x_1, x_2, 0)$. For F : $J_c J_b x = (svx_2 - twx_1, -(su+tw)x_2, t(ux_1 + vx_2))$, then J_a gives $((su+tw)x_2, svx_2 - twx_1, 0)$. Both matched numerically on a basis of P_a for 4000 random triples and exactly (sympy) on six explicit integer triples.
L6. Walk interpretation: F is a 3-edge walk $P_a \rightarrow P_b \rightarrow P_c \rightarrow P_a$, E a 4-edge walk $P_a \rightarrow P_b \rightarrow P_b \rightarrow P_a \rightarrow P_a$	VALID	Composition order is right (rightmost operator applied first). Each intermediate vector lies in the plane of the operator just applied, hence inside $P_a \cup P_b \cup P_c$; each consecutive pair is perpendicular by L0; none is zero by L4. The one step that could look odd, J_b applied to a vector already in P_b , is legitimate — it is the 90° rotation of P_b , and $b \notin P_b$.
L7. $E^n F = [[0, t^{2n}(su+tw)], [-tw, sv]]$ and discriminant (3) $\Delta_n = (sv)^2 - 4t^{2n}tw(su+tw)$	VALID	$E^n = \text{diag}(t^{2n}, 1)$; the product and the char-poly discriminant $\lambda^2 - (sv)\lambda + t^{2n}(su+tw)tw$ are re-derived correctly. Checked numerically against $\text{trace}^2 - 4 \det$ on every random triple (agreement to $1e-9$ relative).
L8. $\Delta_n \rightarrow (sv)^2 > 0$, so some $n \geq 0$ has $\Delta_n > 0$	VALID	$0 < t < 1$ forces $t^{2n} \rightarrow 0$; $s > 0$ and $v \neq 0$ give $(sv)^2 > 0$. The n is <i>effective</i> : if $tw(su+tw) \leq 0$ then $n = 0$ works, otherwise $n = \lceil \log((sv)^2 / (4t \cdot w(su+tw))) / (2 \log t) \rceil$. See Caveats for how large this can get.
L9. $E^n F$ invertible, so the real eigenvalue λ is nonzero	VALID	$\det(E^n F) = t^{2n}(su+tw) \cdot tw \neq 0$ by (1) and $t > 0$. $\Delta_n > 0$ gives two distinct real eigenvalues, hence a real eigenvector $x_0 \neq 0$ in P_a .

step	label	note
L10. Conclusion: closed walk of odd length $3 + 4n$ inside $P_a \cup P_b \cup P_c$, so the induced subgraph is non-bipartite	VALID	The walk ends at $[E \wedge nF \ x_0] = [\lambda x_0] = [x_0]$; working with lines rather than vectors is what makes $\lambda < 0$ harmless, and the writeup is right to phrase it projectively. Length $3 + 4n$ is odd; a closed odd walk in a simple loopless graph contains an odd cycle. Verified in exact arithmetic (below).
C. Corollary: $\mathcal{O}[N_{\mathcal{O}}(S)]$ bipartite $\implies$ $\text{span}(S)$ has dimension ≤ 2	VALID	$N_{\mathcal{O}}([a])$ is precisely the set of lines contained in P_a , so for $[a],[b],[c] \in S$ independent the lemma's subgraph is an <i>induced</i> subgraph of $\mathcal{O}[N_{\mathcal{O}}(S)]$; non-bipartite induced subgraph $\implies$ non-bipartite.
H1. For $h : \mathcal{O} \rightarrow H$ and $S_{\alpha} = h^{-1}\{\alpha\}$, one has $\mathcal{O}[N_{\mathcal{O}}(S_{\alpha})] \rightarrow H[N_H(\alpha)]$	VALID	If $L \sim M$ with $M \in S_{\alpha}$ then $h(L) \sim \alpha$, so $h(L) \in N_H(\alpha)$; and an edge inside $N_{\mathcal{O}}(S_{\alpha})$ maps to an edge inside $N_H(\alpha)$. H locally bipartite $\implies$ target bipartite $\implies$ source bipartite. (Incidentally H has no loops: a loop at α would make $N_H(\alpha) \ni \alpha$ non-bipartite.)
H2. Hence every fibre S_{α} spans dimension ≤ 2	VALID	Direct from C.
H3. Pigeonhole: the $2m+1$ moment-curve lines $[(1,j,j^2)]$, $1 \leq j \leq 2m+1$, with $m = V(H) $	VALID	$\lceil (2m+1)/m \rceil = 3$, so some fibre contains three of them; the Vandermonde determinant $(j-i)(k-i)(k-j) \neq 0$ makes any three independent, contradicting H2. Determinant identity verified symbolically for all triples with $1 \leq i < j < k \leq 25$. The lines are pairwise distinct.
K1. Lemma: $K_{\{p/q\}}$ is locally bipartite when $p < 4q$	VALID	$N(0) = \{q, \dots, p-q\} \subseteq \{q, \dots, 3q-1\}$ because $p - q < 3q$; $A = N(0) \cap \{q, \dots, 2q-1\}$ and $B = N(0) \cap \{2q, \dots, 3q-1\}$ cover it, and any q consecutive residues are pairwise non-adjacent (difference in $[1, q-1] \not\subset [q, p-q]$, using $p \geq 2q$). Vertex-transitivity handles all vertices. Verified exhaustively for $q = 1..12$, $p = 2q..6q$: locally bipartite exactly when $p < 4q$, so the lemma is sharp.
K2. Therefore $\mathcal{O} \not\rightarrow K_{\{p/q\}}$ for $p/q < 4$, i.e. $\chi_c(\mathcal{O}) \geq 4$ (finite-palette definition)	VALID	Immediate from the Theorem + K1. (For $p/q < 2$ the circular clique is not defined, but $\mathcal{O}$ contains triangles so $\chi_c \geq 3$ anyway; not an issue.)
K3. Real-circle definition: scale an r -colouring ($r < 4$) to rational circumference $R \in (r, 4)$, round to a grid of mesh $1/q$	VALID	After scaling, every edge has distance $\geq R/r > 1$. Rounding moves each colour by $\leq 1/(2q)$, hence each distance by $\leq 1/q$ (the writeup's $2/q$ is a safe over-estimate), so distances stay $\geq R/r - 2/q > 1$. Rounded colours live on $p = Rq$ points, and distance ≥ 1 means the $\mathbb{Z}_p$ difference lies in $[q, p-q]$: a homomorphism to $K_{\{p/q\}}$ with $p/q = R < 4$. The estimate is uniform in the vertex, which is exactly why infinitely many vertices are harmless.
U1. Explicit proper 4-colouring with boundary rule ($z > 0$, else $y > 0$, else $x > 0$)	VALID	All representatives have $z \geq 0$, so two same-quadrant non-pole lines have $u \cdot v = (x,y) \cdot (x',y') + zz' > 0$ (half-open quadrant $\implies$ planar angle $< \pi/2 \implies$ positive planar term). Equatorial representatives have argument in $[0, \pi)$, so colour 3 (the interval $[\pi, 3\pi/2)$) is never used on the equator; every other colour-3 line has $z > 0$ and so is not $\perp e_3$. Verified: zero monochromatic edges among 5502 orthogonal pairs on the 865 primitive integer directions of sup-norm ≤ 6 , and on 200 000 random exactly-perpendicular real pairs.
U2. $\chi_c(\mathcal{O}) \leq \chi(\mathcal{O}) \leq 4$, hence $\chi_c(\mathcal{O}) = 4$	VALID	Standard, and the upper bound is anyway given in the problem statement.

step	label	note
W1. Finite witness F_m : the $2m+1$ moment-curve lines plus, for every triple, the lines of its odd walk; $F_m \rightarrow$ any locally bipartite graph on $\leq m$ vertices	VALID	The argument is the finite shadow of H1–H3 and is correct: three initial lines share a fibre, and their odd closed walk lives in $N(S_\alpha)$ and would have to map into a bipartite neighbourhood. Two omissions are cosmetic: triples with two perpendicular normals use the L1 triangle instead, and the least n is not bounded (see Caveats).
W2. Compatibility with the previous attempt (finite subgraphs all have $\chi_c < 4$)	VALID	No contradiction: the palette that realises $\chi_c(F_m) < 4$ has more than m vertices. This is exactly the previous attempt's Proposition 2, and the two statements sit together consistently — see Caveats and Computational check.
W3. “No unproved conjecture, regularity assumption, or computational assertion is used”	VALID	Accurate. The writeup invokes <i>no external theorem at all</i> ; the only facts used are the definitions of χ_c , “odd closed walk $\Rightarrow$ non-bipartite”, “homomorphism into bipartite $\Rightarrow$ bipartite”, and $\chi_c \leq \chi$.

No step was found to be a GAP or an ERROR.

B.3 Reference check

The writeup is genuinely self-contained: it cites no paper, no named theorem, and nothing “from the source paper”. The usual failure mode of a misquoted citation therefore cannot occur here, and `references_ok: true` records that every fact it does use is standard and used correctly (listed in W3 above).

What remains is the **priority check (step 2b)**, which turned up one important document that neither the catalog, nor the OpenProblemGarden entry, nor the earlier referee had located.

B.3.1 The real prior art: Ghebleh’s 2007 thesis

M. Ghebleh, “Theorems and computations in circular colourings of graphs”, Ph.D. thesis, Simon Fraser University, 2007 — <https://summit.sfu.ca/item/8399> (PDF: https://summit.sfu.ca/_flysystem/fedora/sfu_migrate/8399/etd3278.pdf). §4.4, “The Projective Plane Orthogonality Graph”, is exactly this problem, with $\mathcal{O}$ defined as the lines through the origin in $\mathbb{R}^3$ adjacent when perpendicular, and the Kochen–Specker motivation. It contains, with $G_n :=$ the subgraph of $\mathcal{O}$ induced by primitive integer vectors of sup-norm $\leq n$:

- **Theorem 4.37.** $\chi(\mathcal{O}) = 4$ (Kochen–Specker for the lower bound; the 4-colouring is credited to Godsil–Zaks, “Coloring the sphere”, Waterloo technical report, 1988).
- **Proposition 4.39.** $\chi_c(\mathcal{O}) \geq 7/2$, via $\chi_c(G_1) = 7/2$ on the 13-vertex G_1 . *This is the “3.5” that OPG and the catalog record.*
- **Theorem 4.40.** $\chi_c(\mathcal{O}) \geq 11/3$, proved by computer on a 44-vertex, 117-edge subgraph $H_2 \subseteq G_2$ with $\chi_c(H_2) = 11/3$.
- G_3 (145 vertices, 546 edges) admits a (27,7)-colouring, so $11/3 \leq \chi_c(G_3) \leq 27/7 = 4 - 1/7$.
- **Conjecture 4.41:** $\chi_c(\mathcal{O}) = 4$ — the conjecture stated verbatim, a year before the OPG posting.

Three consequences for this review.

1. **The recorded state of the art is stale in the catalog’s favour, not the writeup’s.** The published record lower bound is $11/3 \approx 3.667$, **not 3.5**; and the earlier referee’s own SAT computation ($\chi_c(\mathcal{O}) > 19/5 = 3.8$, and $\geq 23/6$ with the previous attempt’s Proposition 1) already beat *that*. None of this touches the writeup, which claims the full $\chi_c(\mathcal{O}) = 4$ and does not lean on any recorded bound.
2. **The conjecture is published; the theorem is not.** Conjecture 4.41 is the statement being proved, so the writeup is answering a named published conjecture, not a folklore question. The odd-walk lemma, the three-planes construction and the locally-bipartite reduction are **not** in the thesis: §4.4 proceeds entirely by computer search on finite G_n . So this is **not** ALREADY_KNOWN.
3. **Ghebleh’s data cross-validates the whole computational picture.** His G_1, G_2, G_3 are exactly the 13-, 49- and 145-vertex graphs the earlier referee called $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$, with the same edge counts (13/24, 49/138, 145/546 — I rebuilt them). His $\chi_c(H_2) = 11/3$ with $H_2 \subseteq G_2$ agrees with the earlier referee’s independent $\chi_c(\mathcal{O}_2) = 11/3$, and his $\chi_c(G_3) \leq 27/7 = 3.857$ sits correctly above the earlier referee’s $\mathcal{O}_3 \rightarrow K_{15/4}$ (3.75) and $\mathcal{O}_3 \rightarrow K_{11/3}$. Two independent computations, twenty years apart, agree.

B.3.2 Everything else searched

- **OpenProblemGarden entry** (fetched in full): still listed **open**, bibliography section **empty**, last comment 2009 — “I am pretty confident this is still open. Apart from this one many-author paper, I don’t think it has received any significant attention.”
- **arXiv.** 0 results for all: “circular chromatic” AND all: “orthogonality graph”. The 40 most recent abs: “orthogonality graph” AND cat:math.CO papers (2005 $\rightarrow$ 2026-07) contain nothing on $\mathbb{R}^3$ lines or circular colouring (nearest: 2512.01195, quantum chromatic number; 2105.03657, Kunszenti-Kovács–Lovász–Szegedy on random homomorphisms into the orthogonality graph). The 60 most recent abs: “circular chromatic number” AND cat:math.CO papers likewise contain nothing on $\mathcal{O}$.
- **zbMATH, OpenAlex (including its full-text index), Crossref, Semantic Scholar.** No paper proves $\chi_c(\mathcal{O}) = 4$ or improves $11/3$. Ghebleh’s complete 35-item zbMATH list contains nothing on $\mathcal{O}$ — §4.4 was never turned into a paper.

- **Citations of DeVos–Ebrahimi–Ghebleh–Goddyn–Mohar–Naserasr**, “Circular Coloring the Plane”, *SIAM J. Discrete Math.* **21** (2007) 461–465, DOI 10.1137/060664276 (Zbl 1140.05303). Four to five citers: Junosza-Szaniawski $\times 2$ and Chybowska-Sokol–Junosza-Szaniawski–Wesek (all on the *plane* unit-distance graph), a Coxeter-spectral paper, and the thesis itself. The SIAM paper is thesis §4.3 and concerns **only** the plane; its proof is a supremum argument over $\sqrt{3}$ -pairs with a K_4 -e gadget, **not** a locally-bipartite argument, and the obstruction used here could not work there — in the plane unit-distance graph a vertex neighbourhood is a unit circle whose induced graph is a disjoint union of 6-cycles, hence bipartite. Nobody transferred the method to $\mathcal{O}$.
- **Catalog mis-citation (not the writeup’s fault)**. The catalog’s reviewer note guesses DOI 10.1137/050639715, “Coloring an Orthogonality Graph”, as the source of the 3.5 bound. That is Godsil–Newman (arXiv math/0509151) on the ± 1 -hypercube orthogonality graph’s *ordinary* chromatic number. It is not the source; the source is Ghebleh’s thesis §4.4.
- **The §3 lemma as a named result**. K_1 is equivalent to “a graph with a non-bipartite neighbourhood contains an odd wheel, hence $\chi_{-c} \geq 4$ ”. $\chi_{-c}(\text{odd wheel}) = 4$ is standard and citable (Zhu, “Circular chromatic number: a survey”, *Discrete Math.* 229 (2001) 371–410, DOI 10.1016/S0012-365X(00)00217-X), and I re-verified it exactly for W_3, W_5, W_7, W_9 . The explicit general form “ $K_{\lfloor p/q \rfloor}$ is locally bipartite iff $p < 4q$ ” has no located source, but it is a four-line verification the writeup supplies. Note that “locally bipartite” as a term is already in use for an unrelated question (F. Illingworth, “The chromatic profile of locally bipartite graphs”, *JCTB* 156 (2022) 343–388) — a terminology collision, not a mathematical one.
- **Self-contamination check (step 2c)**. Clean. Zero hits anywhere traceable to this campaign: no graph-theory-auto, no gpt-6-astra, no AI-generated writeup; and graph-theory-ai.github.io currently returns GitHub Pages “Site not found”, so the catalog is not even live to be indexed. Every source named above has real authors, a date, and a venue, DOI, arXiv id or institutional-repository record (the thesis is oai:summit.sfu.ca:8399).
- **Limitation to record honestly**. Every general web engine was unusable from this session (the WebSearch budget was exhausted, and Google/Bing/DuckDuckGo/Ecosia/Mojeeek/Startpage/ searx/BASE/CORE all returned a consent wall, CAPTCHA or 403 through WebFetch and curl; the Internet Archive was down). The sweep was therefore run through bibliographic APIs — zbMATH Open, OpenAlex full text, Crossref, Semantic Scholar, arXiv — plus direct fetches of openproblemgarden.org and summit.sfu.ca. For a mathematics priority check that is stronger coverage than a keyword web search, but it would miss an unindexed preprint on a personal page. **No prior art for the theorem or the lemma was found; novelty is unrefuted rather than positively established.**

B.4 Computational check

Scripts (new, this review): verification_astra/scripts/circular_colouring_the_orthogonality_graph/retry_test1_oddwalk.py, retry_test2_locbip.py, retry_test3_exact.py, retry_test4_fourcol.py, retry_test5_witness.py, retry_test6_wheels.py, retry_test7_ghebleh.py. (The test1..test7 scripts in the same directory are the earlier referee’s, for the OPG writeup; I did not rely on them.)

1. **The odd-walk lemma, exact arithmetic** (retry_test3_exact.py). For explicit integer triples the construction is carried out with **no floating point at all**: the matrices E, F are exact integer matrices in the basis $(a \times b, a \times (a \times b))$ of P_a , and the eigenvector and the whole walk live in the real quadratic field $\mathbb{Q}(\sqrt{\Delta_n})$, implemented as exact pairs $(x, y) \leftrightarrow x + y\sqrt{\Delta_n}$ over $\mathbb{Q}$. Every orthogonality, every non-vanishing, every plane-membership and the projective closure are checked as exact identities.

a, b, c	dots (a-b, a-c, b-c)	E	F	n	walk length	result
(1,0,0),(0,1,0),(1,2,3)	(0,1,2)	—	—	—	3	perpendicular case: exact triangle e_1, e_2, e_3
(0,0,1),(1,0,2),(1,1,1)	(2,1,3)	diag(5,4)	$[[1,2],[-3,0]]$	15	63	all checks exact
(1,1,1),(1,2,3),(2,1,5)	(6,8,19)	diag(42,36)	$[[5,48],[-19,0]]$	33	135	all checks exact
(0,0,1),(3,0,4),(1,1,7)	(4,7,31)	diag(25,16)	$[[3,28],[-31,0]]$	14	59	all checks exact
(2,-1,3),(1,4,1),(5,2,-4)	(2,11)	diag(252,1)	$[[-81,2],[-11,0]]$	0	3	exact <i>triangle</i> although a,b,c are pairwise non-orthogonal
(1,0,0),(1,1,0),(1,1,1)	(1,1,2)	diag(2,1)	$[[1,1],[-2,0]]$	4	19	all checks exact
(3,1,2),(1,-2,4),(2,5,1)	(9,13,-4)	diag(294,81)	$[[-41,117],[4,0]]$	0	3	all checks exact

2. **The odd-walk lemma, 4000 random triples** (retry_test1_oddwalk.py). 3988 generic triples fully built (3 skipped for exceeding an n-cap of 4000): **zero failures** of orthogonality, non-vanishing, plane-membership or closure; every walk length odd and equal to $3 + 4n$. Length histogram: 2097 triples give a triangle ($n = 0$), and the tail reaches 9607. A dedicated stress test with a, b nearly parallel ($t = a-b \rightarrow 1$) produced 67 further successes, 0 failures, max walk length 15183.

3. **$K_{\lfloor p/q \rfloor}$ local bipartiteness** (retry_test2_locbip.py). For every $q = 1..12$ and every $p = 2q..6q$, $K_{\lfloor p/q \rfloor}$ is locally bipartite **if and only if** $p < 4q$ — the lemma is correct *and* sharp. The writeup’s explicit A/B bipartition of $N(0)$ was checked to cover $N(0)$ and to consist of independent blocks in every case with $p < 4q$. $K_{\lfloor 7/2 \rfloor}$, $K_{\lfloor 11/3 \rfloor}$, $K_{\lfloor 15/4 \rfloor}$, $K_{\lfloor 19/5 \rfloor}$, $K_{\lfloor 23/6 \rfloor}$ are locally bipartite; K_4 (in any representation $p/q = 4$) and $K_{\lfloor 9/2 \rfloor}$ are not.

4. **Odd wheels** (retry_test6_wheels.py), **cross-checking §3 from the other side**. With an independently written exact χ_{-c} solver (calibrated on $\chi_{-c}(C_5) = 5/2$, $\chi_{-c}(K_4) = 4$, $\chi_{-c}(\text{Petersen}) = 3$, $\chi_{-c}(K_{\lfloor 7/2 \rfloor}) = 7/2$, $\chi_{-c}(K_{\lfloor 11/3 \rfloor}) = 11/3$: $\chi_{-c}(W_3) = \chi_{-c}(W_5) = \chi_{-c}(W_7) = \chi_{-c}(W_9) = 4$, exactly as K_1 requires. Had K_1 been wrong, $W_5 = K_1 + C_5$ would have mapped to $K_{\lfloor 7/2 \rfloor}$; it does not.

5. **The four-colouring** (retry_test4_fourcol.py). On the 865 primitive integer directions of sup-norm ≤ 6 (5502 exactly-orthogonal pairs, integer dot products, no floating point): **0 monochromatic edges**; class sizes 228/228/205/204; the 48 equatorial lines use only colours 0 and 1, and the pole takes colour 2 — exactly the boundary bookkeeping the writeup claims. Also 200 000 random exactly-perpendicular real pairs: 0 monochromatic.

6. **Sharpness of the “finite” hypothesis** (retry_test5_witness.py). $\mathcal{O}$ itself is locally bipartite — $N_{\mathcal{O}}(L)$ is the set of lines of the plane $L^\perp$, in which every line has exactly one perpendicular, i.e. a perfect matching. Confirmed on the 577 integer directions of sup-norm ≤ 5 : every neighbourhood bipartite, maximum degree inside a neighbourhood = 1. Since $\mathcal{O} \rightarrow \mathcal{O}$, the finiteness of H in the Theorem is

essential, and the pigeonhole step H3 is genuinely load-bearing rather than decorative. The writeup states “finite” correctly but does not remark on this.

7. The finite witness F_7 (retry_test5_witness.py). Taking $m = 7$ (the smallest palette, $K_{\{7/2\}}$) and the 15 moment-curve lines $[(1,j,j^2)]$, all $C(15,3) = 455$ triples were processed: 311 walks fully built with **zero** violations (worst numerical residual over all orthogonality / containment / closure checks: 5.3×10^{-15}), lengths from 39 to 11927 (median 639); 144 triples were skipped because the required n exceeds 3000, the largest being $n = 1\,037\,597$, i.e. a single walk of ≈ 4.15 million edges. So F_7 exists and is finite, but with these starting lines $|V(F_7)|$ runs to the order of 10^7 .

8. Consistency with the prior computations. The earlier referee’s SAT results on exact integer orthogonality graphs — $\mathcal{O}_2 \rightarrow K_{\{7/2\}}$, $\mathcal{O}_3 \rightarrow K_{\{11/3\}}$ and $\rightarrow K_{\{15/4\}}$, $\mathcal{O}_4 \rightarrow K_{\{19/5\}}$, while $\mathcal{O}_4 \rightarrow K_4$ — are precisely the finite shadow of this Theorem, and the monotone pattern that referee observed (“the obvious next step is to find a *uniform* construction realising the pattern, i.e. for every k a finite subgraph of $\mathcal{O}$ with no homomorphism to $K_{\{(4k-1)/k\}}$ ”) is exactly what §5 supplies. Nothing in the new proof contradicts that data; it subsumes it. Nor does it contradict the previous attempt’s Proposition 2 (every finite $F \subseteq \mathcal{O}$ has $\chi_c(F) < 4$): F_m does map to some $K_{\{p/q\}}$ with $p/q < 4$, but only for $p > m$. Quantitatively, Proposition 2’s bound $\chi_c(F) \leq 4 - 1/\lfloor (n+1)/4 \rfloor$ gives $\approx 4 - 4 \cdot 10^{-7}$ for F_7 — perfectly compatible with $F_7 \rightarrow K_{\{7/2\}} = 4 - 1/2$, and merely showing that this particular witness is far from vertex-optimal.

9. Cross-check against Ghebleh’s published finite data (retry_test7_ghebleh.py, z3 with a one-hot boolean encoding). I rebuilt his G_n from scratch in exact integer arithmetic (primitive integer directions of sup-norm $\leq n$, adjacency iff the integer dot product is 0) and re-ran the homomorphism questions. The solver was calibrated on eight circular-clique-to-circular-clique instances with known answers, all matching ($K_{\{7/2\}} \rightarrow K_{\{7/2\}}$ sat, $K_{\{11/3\}} \rightarrow K_{\{7/2\}}$ unsat, $K_{\{7/2\}} \rightarrow K_{\{11/3\}}$ sat, $K_4 \rightarrow K_{\{15/4\}}$ unsat, $K_{\{11/3\}} \rightarrow K_{\{27/7\}}$ sat, $K_{\{27/7\}} \rightarrow K_{\{11/3\}}$ unsat, $K_4 \rightarrow K_{\{27/7\}}$ unsat, $K_{\{15/4\}} \rightarrow K_{\{27/7\}}$ sat).

graph	$ V $	$ E $	$\rightarrow K_{\{7/2\}}$	$\rightarrow K_{\{11/3\}}$	$\rightarrow K_{\{15/4\}}$	$\rightarrow K_{\{27/7\}}$	$\rightarrow K_4$
G_1	13	24	sat	sat	sat	sat	sat
G_2	49	138	unsat	sat	sat	sat	sat
G_3	145	546	unsat	unsat	(too slow)	sat	sat

Vertex and edge counts match the thesis exactly (145/546 for G_3). Every sat came with an explicit colouring that I re-verified edge by edge against the $K_{\{p/q\}}$ adjacency rule (0 bad edges in every case). Three things follow. (a) Ghebleh’s Theorem 4.40 chain is reproduced: $\chi_c(G_1) \leq 7/2$ and $\chi_c(G_2) > 7/2$. (b) His (27,7)-colouring of G_3 is confirmed — and it is exactly the kind of object that would *refute* an over-strong reading of the new theorem, so its existence is a real test the theorem passes: G_3 is finite, contains none of the (generally irrational) lines of the odd walks, and maps to a locally bipartite target with 27 vertices. (c) $G_3 \rightarrow K_{\{11/3\}}$ reproduces the earlier referee’s independent result and pushes $\chi_c(G_3)$ strictly above Ghebleh’s $11/3$, so $11/3 < \chi_c(G_3) \leq 27/7$.

No computational check failed.

B.5 Caveats

- Novelty is not established, only unrefuted.** Correctness and priority are separate. I found no prior art (see Reference check) but could not run a general web sweep, and the writeup’s own caveat field says “literature priority has not been checked”. Treat the result as *correct and apparently new*, with novelty unconfirmed.
- The “finite” hypothesis in the Theorem is essential and unremarked.** $\mathcal{O}$ is itself locally bipartite, so “no homomorphism to a locally bipartite graph” is false without finiteness. A reader could misread the Theorem’s headline sentence; the statement itself is correct.
- The least n in the lemma is unbounded.** $\Delta_n > 0$ is guaranteed only asymptotically, and $n \approx \log((sv)^2/(4tw(su+tw)))/(2 \log t)$ blows up as $a \cdot b \rightarrow \pm 1$ (nearly parallel normals) or as $sv \rightarrow 0$. Measured maxima: $n \approx 10^6$ for some moment-curve triples. This is a size, not a soundness, issue — but §5’s “terminating search for n ” and “this construction uses only arithmetic, square roots, and a terminating search” undersell how large F_m becomes ($\approx 10^7$ vertices already for $m = 7$, versus the ≥ 11 vertices that Proposition 2 shows are information-theoretically necessary to block $K_{\{7/2\}}$). The moment curve is a poor choice of starting lines for this purpose; any $2m+1$ pairwise well-separated lines would do.
- §5 forgets the degenerate sub-case.** A triple of the $2m+1$ starting lines with two perpendicular normals must use the L1 triangle rather than the E/F construction. The moment-curve lines do contain such triples in general. Purely cosmetic.
- §3’s rounding constant is loose.** “ $2/q < R/r - 1$ ” with a change of “less than $1/q$ ” per colour is a safe over-estimate (nearest-grid rounding moves a colour by $\leq 1/(2q)$); and “ $R = p/q$, choosing a sufficiently large common multiple of its numerator and denominator” means p/q is no longer in lowest terms, which is harmless since $K_{\{p/q\}}$ is defined for any $p \geq 2q$. Correct as written.
- L2’s normalisation is stated, not derived.** Getting $a = (0,0,1)$, $b = (s,0,t)$ with $s > 0$ and $0 < t < 1$ requires flipping the sign of b when $a \cdot b < 0$ and rotating about a . Both are legitimate and projectively invisible, but the writeup compresses them into “choose their signs and an orthonormal coordinate system”.
- Edge cases checked and harmless.** $m = 0$ (empty H): no homomorphism from a non-empty graph anyway. $p/q < 2$: $K_{\{p/q\}}$ undefined, but $\mathcal{O}$ contains triangles. Loops in H : excluded automatically, since a loop at α makes $H[N_H(\alpha)]$ non-bipartite. Degenerate triples with all three normals mutually orthogonal: covered by L1.
- A natural objection that does not land, worth recording because it looks fatal.** One might argue: “ $K_{\{7/2\}}$ is locally bipartite, yet for its triangle $\{0,2,4\}$ the union $N(0) \cup N(2) \cup N(4)$ is all of $\mathbb{Z}_7$, which is non-bipartite — so an odd walk inside a union of three neighbourhoods cannot exclude a homomorphism to $K_{\{7/2\}}$.” This is a misreading of the argument, and the writeup is not guilty of it. The three lines $[a],[b],[c]$ are not arbitrary: they lie in a **single fibre** $S_\alpha = h^\wedge\{-1\}(\alpha)$, so all three map to the *same* vertex α , and the union of their neighbourhoods maps into the *single* neighbourhood $N_H(\alpha)$ — not into a union of three. That single neighbourhood is bipartite by hypothesis, so the odd walk cannot map into it. The whole force of §2 is that the pigeonhole is applied to fibres, not to triangles. I checked this reading against the text twice: H1 speaks of $S_\alpha = h^\wedge\{-1\}(\alpha)$ throughout, and H3 selects three lines “with the same image under h ”. Relatedly, Ghebleh’s $G_3 \rightarrow K_{\{27/7\}}$ is no obstacle either: G_3 is a 145-line *finite* subgraph and contains none of the (generally irrational) lines of the odd walks.
- What is not claimed, correctly.** The writeup does not claim any single finite subgraph of $\mathcal{O}$ has $\chi_c = 4$, and explicitly says so — which is right, since the previous attempt proves the opposite. The verdict block’s caveats field is honest.

B.6 Referee summary

I set out to find the error and did not find one. The three ways a homomorphism-exclusion argument of this shape normally fails are all closed here. (i) *The parity obstruction established only on a sub-structure*: the odd closed walk is built entirely inside $P_a \cup P_b \cup P_c$, and that union is exactly contained in $N_{\mathcal{O}}(S)$ for a fibre S containing $[a],[b],[c]$ — I re-checked the containment vertex by vertex in exact arithmetic. (ii) *The target's structure misidentified*: “ $K_{\{p/q\}}$ is locally bipartite” is not only true for $p < 4q$ but false for $p \geq 4q$, verified exhaustively, and cross-checked from the other side by computing χ_c of odd wheels. (iii) *Finite and continuous versions conflated mid-proof*: they are not — the whole argument runs in the continuous $\mathcal{O}$, and finiteness is used in exactly one place, the pigeonhole on $|V(H)|$, which is genuinely necessary because $\mathcal{O}$ is itself locally bipartite.

The mechanism, stated in the language the writeup avoids, is a projective holonomy argument: the lines of each plane form an $\mathbb{RP}^1$, the cross-product transitions are projective-linear maps between these circles, an odd cycle is a fixed point of an odd-length composite, and since the 3-step composite F may be elliptic (no real fixed point) one pads it with the 4-step hyperbolic composite E , whose n -th power drives the discriminant to $(sv)^2 > 0$. That padding is what makes the length $3 + 4n$ and keeps the parity odd. It is short, elementary, correct, and I have not seen it elsewhere.

The one genuinely new thing my priority check turned up is that the recorded state of the art is wrong in the *catalog*'s disfavour: the published record is not 7/2 but **11/3**, from Theorem 4.40 of Ghebleh's 2007 SFU thesis, whose §4.4 is devoted to this exact graph and whose Conjecture 4.41 is precisely the statement here proved. That thesis should be the citation in any writeup of this result, and the catalog's 3.5 and its Godsil-Newman mis-citation should both be corrected. It does **not** make the result known: §4.4 is a computer search over finite G_n and contains nothing resembling the odd-walk lemma. It does, however, cross-validate the computational record — his G_1, G_2, G_3 are the earlier referee's $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$ down to the edge counts, and his $\chi_c(H_2) = 11/3$ and $\chi_c(G_3) \leq 27/7$ sit exactly where that referee's independent SAT runs put them.

The tension flagged in my briefing — that the earlier referee's exact-arithmetic SAT computation reached only $\chi_c(\mathcal{O}) > 19/5$ — resolves cleanly in the writeup's favour. That computation bounds χ_c of *finite* subgraphs, which the previous attempt proved can never reach 4; the new theorem is about $\mathcal{O}$ itself, and the finite witnesses it produces (§5) are the uniform family the earlier referee identified as the missing ingredient. $\mathcal{O}_2 \rightarrow K_{\{7/2\}}, \mathcal{O}_3 \rightarrow K_{\{15/4\}}, \mathcal{O}_4 \rightarrow K_{\{19/5\}}$ are exactly the first three instances of the pattern the new proof explains. The two are consistent, and the new result is strictly stronger.

Verdict **CONFIRMED**, high confidence on the mathematics: every step is VALID, no external reference is invoked at all, the interpretation is the intended one, and every finite object in the writeup (the matrices, the walks, the bipartitions, the four-colouring, the witness construction) was verified independently in code — the core lemma in exact arithmetic over a real quadratic field. The single reservation is priority: I found no prior art for either the theorem or the odd-walk lemma across zbMATH, OpenAlex full text, Crossref, Semantic Scholar, arXiv and the thesis literature, but no general web engine was reachable from this session, so novelty should be regarded as unrefuted rather than proven. Subject to that, this is a resolution of the 2008 DeVos–Ghebleh–Goddyn–Mohar–Naserasr problem, and of Ghebleh's Conjecture 4.41: $\chi_c(\mathcal{O}) = 4$.

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